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Research Article

Erinacine A biosynthesis in submerged cultivation of *Hericium erinaceum*: Quantification and improved cultivation

This paper describes the validation of an HPLC method for quantitative determination of erinacine A as well as optimization of the culture medium composition for its effective production in submerged culture of the edible and medicinal mushroom *Hericium erinaceum*. The effects of different medium components were examined by the one-factor-at-a-time method and then by the central composite rotatable design. The most favorable combination of nutrient medium constituents ensuring the highest erinacine A production was found to be: glucose 69.87 g/L, casein peptone 11.17 g/L, NaCl 1.45 g/L, $ZnSO_4$ 55.24 mg/L, KH_2PO_4 1.0 g/L, and pH 4.5. The kinetics of metabolite biosynthesis was examined during the cultivation in a 10-L bioreactor. Under the optimal conditions the biomass yield was 13.3 ± 2.6 g/L, while the production of erinacine A was 192 ± 42 mg/L. The optimal time to obtain the highest production of erinacine A during cultivation of *H. erinaceum* in a bioreactor was eight days.

Keywords: *Hericium erinaceum* / Medium optimization / Method validation / Response surface methodology / Submerged fermentation

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1 Introduction

Hericium erinaceum (Bull.: Fr.) Pers. is a *basidiomycete* mushroom growing on dead deciduous trees. This white rot fungus is widely cultivated in Japan and China and is highly valued for its flavor and various medicinal properties [1, 2]. Among the compounds isolated from fruiting bodies and cultured mycelia of *H. erinaceum*, most interesting are the low-molecular-weight compounds belonging to a group of cyathin diterpenoids (erinacines A–K, P, and Q). Several of them, *i.e.* erinacines A–H, are known to have a potent stimulating effect on nerve growth factor (NGF) synthesis *in vitro* [3–9]. The NGF synthesis stimulators can be applied as potential drugs against central nervous system disorders due to degeneration of neurons, such as Alzheimer's disease [10, 11]. Erinacine A, the main representative of this group of compounds, has a strong enhancing effect on NGF synthesis, much stronger than epinephrine [3]. Furthermore, this compound increases catecholamine and NGF content in the central nervous system of rats [12].

Unfortunately, the fruiting bodies of *H. erinaceum* may vary considerably in composition depending on environmental conditions and site of their collection, and their traditional cultivation takes a long time [13]. Therefore it seems highly desirable to cultivate *H. erinaceum* in laboratory conditions under constantly controlled culture parameters. In this way a high yield of mycelium of constant composition can be obtained, simultaneously affording high concentrations of bioactive metabolites, useful in formulation of nutraceuticals and functional foods [14–17]. The biosynthesis of erinacine A in submerged culture could not only considerably lower the costs of its production but is also practically the only effective way to obtain this compound, because the chemical synthesis of cyathin diterpenoids, although not impossible, is a protracted multi-stage process affording a very low yield [18].

In our previous papers we described the optimization of culture medium composition in *H. erinaceum* submerged cultivation for the production of polysaccharides [19, 20]. In this work we investigated the effect of culture medium composition on biosynthesis of erinacine A by means of central composite rotatable design (CCRD) experiment and response surface methodology (RSM) analysis. The CCRD method allows several factors to be examined simultaneously

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Abbreviations: ANOVA, analysis of variance; **CCRD**, central composite rotatable design; **CSP**, corn steep powder; **LOD**, limit of detection; **LOQ**, limit of quantitation; **NGF**, nerve growth factor; **RSM**, response surface methodology; $\mathbf{R_t}$, retention time; **SD**, standard deviation; **SE**, standard error; **%RSD**, relative standard deviation

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and guarantees the choice of optimal conditions for the bioprocess. Moreover, by use of the CCRD method the number of experiments can be significantly reduced and mutual relationships between parameters found [21, 22]. In the last stage of the experiment the *H. erinaceum* mycelium was cultivated in a bioreactor in culture medium with the optimized composition. The kinetics of erinacine A biosynthesis was investigated and the time necessary to obtain its maximal yield was determined.

This is the first report on the specific quantitative method for the determination of erinacine A, as well as on the optimization of submerged culture conditions in *H. erinaceum* liquid culture for the production of a low-molecular-weight secondary metabolite.

2 Materials and methods

2.1 HPLC method for quantitative determination of erinacine A; calibration curve

Erinacine A used as a standard was obtained from the mycelium of *H. erinaceum* according to the method described earlier [23] and was re-purified. The structure and purity (>98.0%) were confirmed by ^{1}H, ^{13}C, HETCOR (heteronuclear correlation), and DEPT (distortionless enhancement by polarization transfer) NMR spectroscopy (400 MHz), and by HR-ESI-MS (high-resolution electrospray ionisation mass spectroscopy; ion peak *m/z* 455.24264, required 455.24041). A stock solution of erinacine A (1.00 mg/mL) was prepared in methanol. Standard solutions with the final concentration range of 2–500 μg/mL were obtained by proper dilution of stock solution, with triplicate determination at each level. The data of peak areas plotted against the corresponding concentrations were treated by linear least-square regression analysis. The homoscedascicity test, the lack of fit test, and the zero intercept test were performed to verify the linearity of the calibration curve [24].

The samples were injected onto an HPLC column Supelcosil LC-18-DB, 5 μm particle size, 25 cm × 4.6 mm (Supelco, USA). The analysis was carried out with an HPLC instrument equipped with a 20 μL sample loop, an LC-10 Atvp pump, an SCL-10 Avp controller, and an SPD-10 Avp UV detector (Shimadzu, USA) at 340 nm (in the range of absorption maximum wavelength; data not shown). Separation was performed at 40°C using methanol–water 70:30 v/v isocratic elution at 1.0 mL/min. Retention time of erinacine A was 6.33 ± 0.12 min. All solvents used in the mobile phases were of HPLC grade, supplied by Merck (Darmstadt, Germany).

Validation of the HPLC method was carried out according to the International Conference on Harmonization guideline ICH, Q2A (R1) (Validation of analytical procedures: Text and methodology, 2005, http://www.ich.org/LOB/media/MEDIA417.pdf).

2.2 Precision and accuracy

Repeatability of the sample application and measurement of peak area was carried out for six replicates of same concentration (10.0 μg/mL of erinacine A). Each sample was prepared separately and measured at 340 nm. For method precision, the intra- and inter-day variation for determination of erinacine A ($n = 3$) was performed with three different concentration levels of 10.0, 20.0, and 30.0 μg/mL. The inter-day precision (intermediate precision) was studied by comparing the analysis performed on three different days, whereas assay for the intra-day variation was performed with 3-h intervals. The precision of the method was expressed in terms of %RSD (relative standard deviation) and standard error (SE) of peak area. The accuracy of the method was expressed as percent recovery (100% × mean back-calculated concentration/theoretical concentration).

The accuracy and precision of the extraction procedure of erinacine A from biological material were checked as follows: nine samples of *H. erinaceum* mycelium (each of 1.3 g) were extracted as described in Section 2.8 and erinacine A content was measured. Recovery studies were carried out by applying the method to the pre-analyzed samples of *H. erinaceum* extract to which known amount of erinacine A corresponding to 50, 100, and 150% of the initial erinacine A content had been added.

2.3 LOD and LOQ

The LOD (limit of detection) and LOQ (limit of quantitation) were estimated by the repeated analysis ($n = 6$) of the blank following the same method as explained in Section 2.1. The SD (standard deviation) of the response was determined, and the LOD was expressed as 3.3 × SD/slope of the calibration curve of erinacine A, whereas LOQ was expressed as 10 × SD/slope of the calibration curve of erinacine A.

2.4 Specificity

The specificity of erinacine A determination with regard to co-eluting components of *H. erinaceum* extract was investigated by analyzing and comparing the R_t (retention time) values and UV-spectra for erinacine A in sample with that for standard.

2.5 Microorganism, preparation of inocula, and flask culture

H. erinaceum (Bull.: Fr.) Pers. mycelium was maintained and cultured as described previously [23]. The effect of different carbon sources, nitrogen sources, mineral elements, and culture medium pH on erinacine A production, as well as on mycelial growth, was examined. Cultivations were in 500-mL Erlenmeyer flasks containing 200 mL of a basal nutrient medium composed of glucose 50.0 g/L (POCh SA, Gliwice, Poland), yeast extract 10.0 g/L, casein peptone 10.0 g/L (Biocorp Sp. z o.o., Warsaw, Poland), KH_2PO_4 1.0 g/L (POCh SA, Gliwice, Poland), and pH 5.0 on a rotary shaker (New Brunswick, USA) at 110 rpm at 25°C. The pH value of the culture broth was determined with a digital pH meter (inoLab pH Level 1, WWT, Germany).

To chose the best carbon source for erinacine A production, the following carbon sources at a concentration of 50.0 g/L

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were added to the basal medium instead of glucose: fructose, lactose, starch, sucrose, xylose (POCh, Gliwice, Poland), mannose, trehalose, and malt extract (Sigma-Aldrich, Germany). The effect of nitrogen source on erinacine A production was examined in a basal medium in which the tested nitrogen source (2.8 g/L calculated as N) was substituted for yeast extract and casein peptone. We tested both inorganic nitrogen sources such as ammonium sulfate and potassium and ammonium nitrates, and organic ones such as urea, casein peptone, yeast extract, soy peptone (Biocorp Sp. z o.o., Warsaw, Poland), and corn steep powder (CSP). The effect of initial pH of culture medium was tested within the range from 4.0 to 7.0 (4.0, 4.5, 5.0, 5.5, 6.0, and 7.0). The inorganic salts, NaCl, $MgSO_4 \cdot 7H_2O$, $ZnSO_4 \cdot 7H_2O$, and $CuSO_4 \cdot 5H_2O$ (POCh, Gliwice, Poland), were added to the basal medium and tested at the concentrations of 1.0, 0.5, 0.05, and 0.005 g/L, respectively.

After 28 days the cultivation was terminated, and the mycelial dry mass and erinacine A content in mycelium were examined. This time of cultivation was chosen on the basis of the literature data [3–5, 9]. The culture components found to be most effective in the one-factor-at-a-time method were then tested by the CCRD method. The application of this technique allowed us to choose the optimal concentrations of the components, which resulted in the high production of erinacine A. At least two replications were performed for each cultivation to ensure reproducibility.

2.6 Experimental design and data analysis

The RSM used in the present study is a full 2^4 factorial CCRD, which contained eight star points ($\alpha = \pm 2$) in two replicates and six replications at the central points ($n_0 = 6$). A total number of 54 shake-flasks experiments were required to optimize the chosen variables. The following equation was used for coding the test variables:

$$X_i = \frac{x_i - x_i^0}{\Delta x_i} \quad (1)$$

where X_i is the dimensionless coded value of the i-th test variable, x_i the real value of the i-th test variable, x_i^0 the real value of the i-th test variable at the center point, and Δx_i step change value.

The relationship of the nutrient medium components (independent variables) and the mycelial growth, and erinacine A production (the response) was fitted to a predictive second-order polynomial equation:

$$Y_i = \beta_0 + \sum_{i=1}^{k} \beta_i X_i + \sum_{i=1}^{k} \beta_{ii} X_i^2 + \sum_{i<j}^{k} \sum \beta_{ij} X_i X_j \quad (2)$$

where Y_i is the predicted response, subscripts i and j take values from 1 to the number of variables, β_0 is a constant, β_i the linear coefficient, β_{ii} the quadratic coefficient, β_{ij} the cross-product coefficient, k number of factors, and X_i and X_j are the coded dimensionless values of the investigated variables. The softwares STATISTICA 8 (StatSoft, USA) and Design-Expert 7 (Stat-Ease, USA) were used for experimental design, the analysis of variance (ANOVA), and graphical analysis of the data. The statistical significance of the quadratic model was expressed by an F-test and the quality of its fit was evaluated by the coefficient of determination R^2. The significance of the regression coefficients was tested by a t-test and the p-values were used as a tool to check the significance of each coefficient.

2.7 Bioreactor fermentation

The culture medium of optimal composition for the biosynthesis of erinacine A, selected by the CCRD method, was used to investigate the kinetics of the bioprocess. A 10-L bioreactor (Biotec, UK) was used, with aeration of 1 vvm and agitation speed 150 rpm. The fermentation was for 28 days at 25°C. The cultivation parameters such as the temperature, pH, and agitation speed were monitored during the bioprocess. The samples collected at regular time intervals were analyzed for dry mass content and erinacine A concentration. In the samples of culture broth collected during the cultivation of *H. erinaceum* in the bioreactor the residual sugar concentration was assayed by the 3,5-dinitrosalicylic acid (Sigma-Aldrich) method [25] with a UVmini 1240 Shimadzu (Japan) spectrophotometer.

The specific growth rate μ (1/day) was calculated according to the equation $\mu = (1/X)(dX/dt)$, where X is cell concentration (g/L) at time t (day). The specific consumption rate of substrate (g/(g × day)) was calculated according to the equation $Q_{S/X} = (1/X)(dS/dt)$, where S is the concentration of carbon source (g/L) at time t (day). The specific production rate of metabolite (mg/(g × day)) was calculated according to the equation $P_{P/X} = (dP/dt)(1/X)$, where P is the concentration of metabolite (mg/L) at time t (day). The yield of metabolite on substrate (mg/g) was calculated from the formula $Y_{P/S} = (dP/dt)/(dS/dt)$.

2.8 Analytical procedure for the determination of cell dry weight and erinacine A content in the mycelium

When the cultivation of *H. erinaceum* was finished the fungal biomass was filtered through filter paper (Whatman no. 2) under reduced pressure, and then washed thoroughly with 250 mL of distilled water. The mycelium was dried at 40°C, powdered, subsequently sieved using an 0.150 mm stainless steel mesh sieve, and the dry mass was measured at 105°C with a moisture analyzer (Sartorius, Germany). The samples of powdered mycelium (1.3 g) obtained from both shake-flask cultures and the bioreactor were then submitted to fractional extraction in a Soxhlet apparatus. Successive extraction with 50 mL of petroleum ether and 50 mL of 96% ethanol gave two fractions of compounds with different polarities. Solvents used in the extraction procedure were purchased from POCh (Gliwice, Poland) and were glass-distilled before use. The ethanol extract obtained (*ca.* 50 mL) was evaporated to dryness under reduced pressure and the dry residue dissolved in 5.0 mL of methanol. The sample was injected onto an HPLC column as described in Section 2.1. The content of erinacine A was calculated on the basis of a standard curve.

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2.9 Statistical analysis

All results are expressed as the mean of at least two replicates ± SD. The obtained data were subjected to a one-way ANOVA with the aid of the STATISTICA 8 software. Differences between means at $p < 0.05$ level were considered significant.

3 Results and discussion

3.1 Calibration curve of erinacine A

The representative HPLC chromatogram of the extract of *H. erinaceum* mycelium is shown in Fig. 1. The linear regression for the calibration curves ($n = 3$, $p = 9$) showed a correlation coefficient (r) of 0.9994 with respect to the peak area, demonstrating a good linear relationship in the range of 2–50 μg/mL (above this concentration a non-linear response was observed). No significant difference was observed in the slopes of standard curves. To verify the linearity of the model additional statistical tests were performed. The model passed the homoscedasticity test (Cochran's test), the lack of fit test, and the zero intercept test (Table 1).

3.2 Method validation

The %RSD for repeatability of sample application (10.0 μg/mL) and measurement of peak areas were found to be 0.84 and 1.12%, respectively. The LOD and LOQ based on the standard deviation of the response and the slope were found to be 0.228 and 0.691 μg/mL, respectively, indicating high sensitivity of the proposed method (Table 1).

The low value of %RSD (< 3.1%) and SE revealed in the course of inter- and intra-day variation indicated good precision of the developed HPLC method. All tested values lie within the 95% confidence interval, which suggest that the method is sufficiently accurate (Table 2).

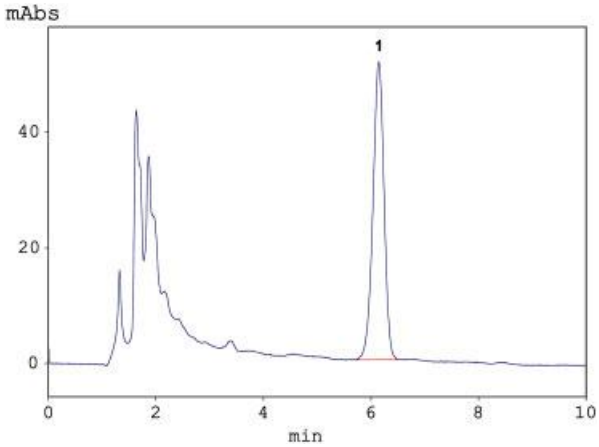

Figure 1. Representative HPLC chromatogram of the extract of *H. erinaceum* mycelium (UV detection an 340 nm). Peak 1 refers to erinacine A.

Table 1. Statistical data for the calibration curve of erinacine A.

Parameter	Values
Linearity range	2–50 μg/mL
Correlation coefficient	0.9994
Slope ± SD	$8.86 \times 10^3 \pm 0.3 \times 10^3$
Confidence limit of slope	8.58×10^3–9.14×10^3
SE of slope	0.12×10^3
Intercept ± SD	$-7.20 \times 10^3 \pm 0.32 \times 10^3$
Confidence limit of intercept	$-14.24 \times 10^3 - (-0.18 \times 10^3)$
SE of intercept	2.97×10^3
C for Cochran's test[a)]	0.393
F for the lack of fit test[b)]	2.571
p for the zero intercept test	0.0456
LOD	0.228 μg/mL
LOQ	0.691 μg/mL

[a)] The critical value of $C_{3,9}$ at 95% confidence interval is 0.478.
[b)] The critical value of $F_{0.05,7,18}$ at 95% confidence interval is 3.467.

Table 2. Precision and accuracy of HPLC method ($n = 6$).

Concentration (μg/mL)	Precision[a)]			Accuracy[b)] (%)
	SD	%RSD	SE	
Intra-day				
10.0	0.776×10^3	0.96	0.317×10^3	99.77
20.0	4.71×10^3	2.78	1.92×10^3	99.60
30.0	5.57×10^3	2.14	2.27×10^3	100.40
Inter-day				
10.0	1.02×10^3	1.26	0.418×10^3	98.11
20.0	3.31×10^3	1.94	1.35×10^3	100.35
30.0	8.04×10^3	3.07	3.28×10^3	101.33

[a)] Precision as SD, %RSD, and SE of peak area
[b)] Accuracy = mean back calculated concentration/theoretical concentration × 100%

Table 3. Effect of carbon source on erinacine A content and production and mycelial growth in shake-flask cultures of *H. erinaceum*.[a)]

	Erinacine A production (mg/L)	Erinacine A content (mg/g)	Mycelial growth (g/L)
Control	0.0374 ± 0.0022	0.0165 ± 0.0031	2.30 ± 0.35
Glucose	1.18 ± 0.22*	0.106 ± 0.015*	11.04 ± 0.71*
Fructose	0.210 ± 0.042*	0.017 ± 0.001	12.5 ± 2.3*
Lactose	0.260 ± 0.025*	0.0283 ± 0.0033*	9.20 ± 0.43*
Malt extract	0.21 ± 0.04*	0.018 ± 0.003	11.39 ± 0.32*
Mannose	0.15 ± 0.03*	0.0151 ± 0.0026	9.9 ± 0.2*
Starch	0.87 ± 0.17*	0.06 ± 0.01*	13.6 ± 2.6*
Sucrose	0.0395 ± 0.0025	0.0124 ± 0.0008	3.2 ± 0.4*
Trehalose	0.97 ± 0.09*	0.11 ± 0.01*	8.69 ± 0.05*
Xylose	0.14 ± 0.01*	0.0136 ± 0.0011	10.7 ± 0.3*

* Significant difference *versus* control at the 5% significance level.
[a)] The carbon sources were tested at a concentration of 50.0 g/L. All experimental data are values of mean ± SD of triplicate determinations.

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The sample solution showed several additional peaks at R_t values from 1.3 to 3.4 min, which were significantly less intense that the principal peak of erinacine A (R_t 6.33±0.12 min) (Fig. 1). All impurities were well-resolved and did not interfere with the analysis of erinacine A under the optimized chromatographic conditions. The selectivity factor (α) and the resolution factor (R_S) measured for the erinacine A peak from the nearest resolving peak in *H. erinaceum* extract were 1.84±0.02 and 5.8±0.2, respectively, which indicates that the proposed method is very specific.

When optimizing the extraction procedure, no erinacine A in the *n*-hexane extract was found. The maximum content of

Table 4. Effect of nitrogen source on mycelial growth and erinacine A content and production in shake-flask cultures of *H. erinaceum*.[a)]

	Erinacine A production(mg/L)	Erinacine A content(mg/g)	Mycelial growth(g/L)
Control	1.25±0.04	0.404±0.022	3.09±0.26
Ammonium nitrate	19±3*	6.19±0.16*	3.10±0.52
Ammonium sulfate	17.6±3.1*	6.82±0.44*	2.57±0.34
Casein peptone	27±4*	3.78±0.43*	7.2±0.5*
CSP	17.0±2.4*	1.32±0.18*	13±1*
Potassium nitrate	0.42±0.04*	0.191±0.024*	2.2±0.1*
Soy peptone	3.1±0.6*	0.31±0.06	9.9±0.9*
Urea	0.010±0.001*	0.00826±0.00021*	1.20±0.09*
Yeast extract	1.09±0.18	0.114±0.015*	9.6±0.6*

* Significant difference *versus* control at the 5% significance level.

a) The effect of nitrogen source was examined in a basic medium in which the tested nitrogen source (2.8 g/L calculated as N) was substituted for yeast extract and casein peptone. All experimental data are values of mean±SD of triplicate determinations.

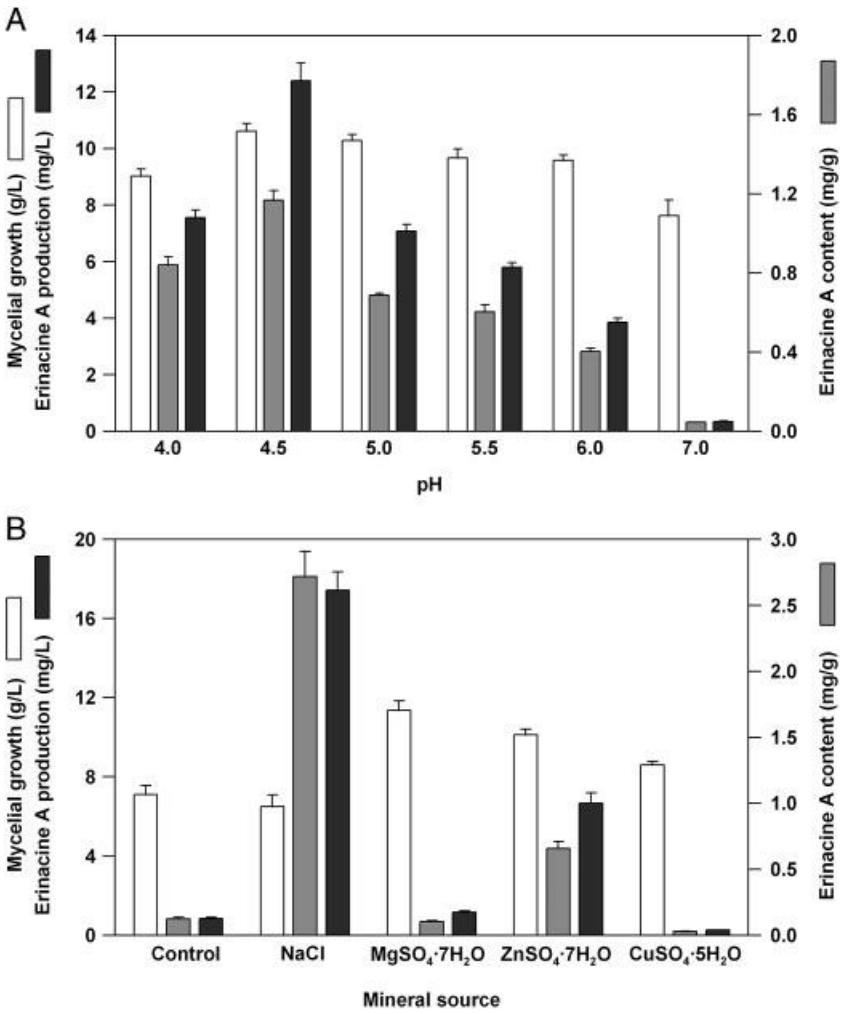

Figure 2. . Effect of (A) initial pH of culture medium and (B) mineral elements on mycelial growth and erinacine A production in shake-flask cultures of *H. erinaceum*. Mineral elements were tested at the following concentrations: NaCl 1.00 g/L, $MgSO_4 \cdot 7H_2O$ 0.50 g/L, $ZnSO_4 \cdot 7H_2O$ 50.0 mg/L, and $CuSO_4 \cdot 5H_2O$ 5.0 mg/L. All experimental data are values of mean ± SD of triplicate determinations. Control refers to the nutrient medium with no minerals added.

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erinacine A in the ethanol extract was achieved after 3 h of extraction. Repetitive measurements of erinacine A content in nine samples of individual mycelial extract showed appropriate precision of the method (%RSD < 3.0). The developed method when used for the determination of erinacine A in *H. erinaceum* extract after spiking with 50, 100, and 150% of the additional erinacine A afforded mean recovery of 101.4, 100.4, and 99.4%, respectively. The %RSD (with SE) values after spiking with 50, 100, and 150% of extra erinacine A were found to be 1.19 (0.70), 1.57 (0.91), and 2.11 (1.21), respectively, confirming accuracy of the extraction procedure as well as of the method of erinacine A determination in biological samples.

3.3 Effect of nutrient medium components and initial pH value

The well-planned CCRD experiment requires optimal variable selection, which could be previously achieved by use of the one-factor-at-a-time method. We have investigated the effect of carbon source, nitrogen source, mineral elements, and initial pH of the culture medium on erinacine A production and on mycelial growth of *H. erinaceum*. Compared with our previous paper [20] no statistically important differences between mycelial growth after three and four weeks of cultivation were noticed when the same nutrient medium components or initial nutrient medium pH were applied. Insignificant depletion of biomass yield after 4 wk resulted from a gradual lysis of mycelial cells. The longer time of cultivation for the production of erinacine A was chosen on the strength of the literature data, which confirm that the biosynthesis of low-molecular-weight secondary metabolites proceeds more intensively in the late stationary phase after 28 days of cultivation [3–5, 9, 26]. Detailed results concerning the effect of carbon source on mycelial growth as well as the content and production of erinacine A are presented in Table 3. It has often been reported that the effects of different carbon sources on mycelial growth are not closely correlated with those on secondary metabolite biosynthesis. This phenomenon is due to a different effect of catabolic repression on secondary metabolism [26, 27]. In our investigations we have shown that in submerged cultivation of *H. erinaceum*, as in other mycelial cultures of higher fungi, the content of secondary metabolite (mg/L) in the fermentation broth depends not only on its concentration in the mycelium (mg/g) but also on biomass yield (g/L). Therefore the optimal composition of culture medium for mycelial growth can differ from the composition suitable for effective production of a secondary metabolite. Although the highest biomass yield was obtained when the mycelium grown in the medium contained starch and fructose, the highest erinacine A production was obtained in the medium contained glucose. We chose glucose for further medium optimization in the CCRD experiment, since the addition of this sugar afforded a very high production of erinacine A. Glucose has been usually used as a suitable carbon source in submerged cultivations of many fungi such as *Xerocomus badius*, *Agrocybe cylindracea*, or *Collybia maculata* [22, 27, 28].

Table 4 presents the results of the effect of nitrogen sources on mycelial growth and production of erinacine A. Our experiments showed that the most favorable substrates for biomass growth were complex nitrogen sources, of which CSP had the greatest positive effect. The obtained data corroborated a well-known thesis that higher fungi choose for their growth complex nitrogen sources, which are rich in essential amino acids [22, 29]. Simple nitrogen sources, such as inorganic salts and urea, clearly inhibited growth of mycelium, comparably to that in the control medium with no nitrogen source added. However, the erinacine A content was very high in those culture media. It is interesting to note that biosynthesis of erinacine A proceeded with different intensities when various nitrogen sources were used, which is consistent with the current knowledge that different nitrogen compounds demonstrate different effects on cell growth and secondary metabolite biosynthesis [30]. For further investigations by the CCRD method we selected casein peptone, because with this nitrogen source we obtained the highest erinacine A production with coincidental high biomass yield.

According to the results of our experiments, a more acidic culture medium pH (*i.e.* pH 4.5) positively affects both mycelial growth and metabolite production (Fig. 2A). Similar observations were made by many authors in cultivations of higher fungi such as *Cordyceps militaris* or *Ganoderma lucidum* [16, 30].

Examining the effect of mineral elements the highest content of erinacine A in mycelium was found in the presence of NaCl and was about 22 times as great as in the control medium. Similarly, a positive effect of NaCl was found on erinacine A production (Fig. 2B). We observed rather surprising effects of copper ions, which other authors had used to increase the biosynthesis of secondary metabolites. In our experiment both the content and the production of erinacine A were considerably lower than in control medium with no addition of that element. The possible

Table 5. Actual and coded values of the independent variables tested in CCRD.

Variables	Symbol		Coded levels				
	Uncoded	Coded	−2	−1	0	1	2
Glucose (g/L)	x_1	X_1	30.0	40.0	50.0	60.0	70.0
Casein peptone (g/L)	x_2	X_2	10.30	15.45	20.60	25.75	30.90
NaCl (g/L)	x_3	X_3	0.50	0.75	1.00	1.25	1.50
$ZnSO_4 \cdot 7H_2O$ (mg/L)	x_4	X_4	25.0	37.5	50.0	62.5	75.0

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explanation of this phenomenon could be a rapid transformation of erinacine A into another cyantan metabolite [8]. For further investigations in the CCRD experiment we selected NaCl because it strongly enhanced the biosynthesis of erinacine A while mycelial growth was uninhibited. Another component chosen for the CCRD experiment was $ZnSO_4 \cdot 7H_2O$. This mineral compound was found to considerably enhance mycelial growth and also to increase erinacine A production.

3.4 Optimization of culture conditions with CCRD and RSM

The CCRD is one of the mathematical methods of experiment projecting used to investigate the relationships between various components of a culture medium and to determine their effect on mycelial growth and production of specific metabolites. The range and the levels of the independent variables tested in the CCRD experiment are tabulated (Table 5). In this work we applied the full-factorial 2^4 CCRD, which in the present case means that in 54 experimental runs four variables (*i.e.* different components of culture medium) were examined at five different concentrations. A big advantage of this method is considerable minimization of the number of experiments in comparison with the classical one-factor-at-a-time method; thanks to it the optimal composition of the culture medium can be established over a shorter time. The usefulness of this method has been proved by other authors who investigated the optimization of culture conditions of mushrooms for the production of secondary metabolites [22, 31, 32].

For the CCRD experiment glucose was chosen as the carbon source and casein peptone as the nitrogen source. These medium components ensured very high erinacine A production with simultaneous high mycelial growth. The other components selected for the experiment were NaCl, which enhanced erinacine A production, and also $ZnSO_4 \cdot 7H_2O$, which additionally enhanced mycelial growth. The other culture conditions were as follows: temperature 25°C, agitation speed 110 rpm, time course of cultivation 28 days. The pH of culture medium was 4.5 and was selected as optimal on the basis of the one-factor-at-a-time experiment. The concentration of KH_2PO_4 was fixed at 1.0 g/L.

The design matrix for the CCRD experiment, together with the experimental results and the predicted responses of mycelial growth and erinacine A production are given in Table 6. The experimental values obtained from the CCRD were regressed by a quadratic polynomial equation, where only the

Table 6. CCRD matrix of independent variables and the corresponding experimental and predicted values of mycelial growth and erinacine A production by submerged culture of *H. erinaceum*.

Run	Factor[a)]				Mycelial growth (g/L)		Erinacine A (mg/L)	
	X_1 (g/L)	X_2 (g/L)	X_3 (g/L)	X_4 (mg/L)	Observed[b)]	Predicted	Observed[b)]	Predicted
1	40.0 (−1)	15.45 (−1)	0.75 (−1)	37.5 (−1)	9.26 ± 0.14	9.65	28.40 ± 0.42	27.97
2	40.0 (−1)	15.45 (−1)	0.75 (−1)	62.5 (1)	10.50 ± 0.15	10.29	29.75 ± 0.42	29.38
3	40.0 (−1)	15.45 (−1)	1.25 (1)	37.5 (−1)	10.71 ± 0.25	10.52	31.5 ± 0.45	31.23
4	40.0 (−1)	15.45 (−1)	1.25 (1)	62.5 (1)	11.30 ± 0.36	11.35	32.1 ± 0.54	32.13
5	40.0 (−1)	25.75 (1)	0.75 (−1)	37.5 (−1)	11.40 ± 0.22	11.07	25.0 ± 0.7	24.54
6	40.0 (−1)	25.75 (1)	0.75 (−1)	62.5 (1)	11.13 ± 0.42	11.26	26 ± 1	25.78
7	40.0 (−1)	25.75 (1)	1.25 (1)	37.5 (−1)	11.3 ± 0.3	11.33	29.05 ± 0.36	29.03
8	40.0 (−1)	25.75 (1)	1.25 (1)	62.5 (1)	11.7 ± 0.7	11.71	29.9 ± 0.6	29.76
9	60.0 (1)	15.45 (−1)	0.75 (−1)	37.5 (−1)	12.01 ± 0.17	11.96	30.88 ± 0.54	30.51
10	60.0 (1)	15.45 (−1)	0.75 (−1)	62.5 (1)	12.13 ± 0.54	12.26	31.5 ± 0.6	31.25
11	60.0 (1)	15.45 (−1)	1.25 (1)	37.5 (−1)	12.08 ± 0.18	12.17	34 ± 1	33.89
12	60.0 (1)	15.45 (−1)	1.25 (1)	62.5 (1)	12.38 ± 0.33	12.66	34.10 ± 0.71	34.11
13	60.0 (1)	25.75 (1)	0.75 (−1)	37.5 (−1)	12.53 ± 0.63	12.67	28.50 ± 0.43	28.18
14	60.0 (1)	25.75 (1)	0.75 (−1)	62.5 (1)	12 ± 1	12.53	29.0 ± 0.6	28.74
15	60.0 (1)	25.75 (1)	1.25 (1)	37.5 (−1)	12.1 ± 0.2	12.27	32.9 ± 0.5	32.79
16	60.0 (1)	25.75 (1)	1.25 (1)	62.5 (1)	12.53 ± 0.21	12.32	32.7 ± 0.8	32.83
17	30.0 (−2)	20.60 (0)	1.00 (0)	50.0 (0)	9.54 ± 0.11	9.67	28.7 ± 0.6	29.11
18	70.0 (2)	20.60 (0)	1.00 (0)	50.0 (0)	12.9 ± 0.4	12.59	34.5 ± 0.5	34.73
19	50.0 (0)	10.30 (−2)	1.00 (0)	50.0 (0)	11.02 ± 0.32	10.84	35.55 ± 0.36	36.04
20	50.0 (0)	30.90 (2)	1.00 (0)	50.0 (0)	11.9 ± 0.4	11.92	31.1 ± 0.3	31.34
21	50.0 (0)	20.60 (0)	0.50 (−2)	50.0 (0)	11.57 ± 0.11	11.42	30.1 ± 1.2	31.05
22	50.0 (0)	20.60 (0)	1.50 (2)	50.0 (0)	12.08 ± 0.55	12.08	38.6 ± 0.9	38.41
23	50.0 (0)	20.60 (0)	1.00 (0)	25.0 (−2)	11.93 ± 0.18	11.90	18.8 ± 0.6	19.46
24	50.0 (0)	20.60 (0)	1.00 (0)	75.0 (2)	12.70 ± 0.21	12.58	20.9 ± 0.5	20.92
25	50.0 (0)	20.60 (0)	1.00 (0)	50.0 (0)	12.47 ± 0.14	12.48	32.18 ± 0.18	32.58
26	50.0 (0)	20.60 (0)	1.00 (0)	50.0 (0)	12.4 ± 0.2	12.48	32.80 ± 0.15	32.58
27	50.0 (0)	20.60 (0)	1.00 (0)	50.0 (0)	12.5 ± 0.3	12.48	32.78 ± 0.61	32.58

[a)] X_1 – glucose, X_2 – casein peptone, X_3 – NaCl, X_4 – $ZnSO_4 \cdot 7H_2O$; values in parentheses are coded variables.
[b)] Results are reported as mean ± SD of two determinations.

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significant terms were considered ($p<0.05$), giving two empirical correlations:

$$Y_1 = 12.48 + 0.73X_1 + 0.27X_2 + 0.16X_3 + 0.17X_4 - 0.34X_1^2 - 0.28X_2^2 - 0.18X_3^2 - 0.18X_1X_2 - 0.16X_1X_3 - 0.15X_2X_3 \quad (3)$$

$$Y_2 = 32.58 + 1.41X_1 - 1.18X_2 + 1.84X_3 + 0.37X_4 + 0.27X_2^2 + 0.55X_3^2 - 3.10X_4^2 + 0.27X_1X_2 - 0.15X_2X_3 \quad (4)$$

where Y_1 is the mycelial growth (g/L), Y_2 is the erinacine A production (mg/L), and X_1, X_2, X_3, and X_4 are coded values of the independent variables (glucose, casein peptone, NaCl, and $ZnSO_4 \cdot 7H_2O$, respectively).

The parameter estimates of the predicted response along with the corresponding *t*-values and SE are shown in Table 7. Parameter estimates indicate that glucose had the largest effect on mycelial growth, followed by casein peptone, $ZnSO_4 \cdot 7H_2O$, and NaCl. The order of effects on erinacine A production was as follows: NaCl > glucose > $ZnSO_4 \cdot 7H_2O$ > casein peptone. The fit of the models was expressed by the coefficient of determination R^2, which was found to be 0.8894 and 0.9827 for mycelial growth and erinacine A production, respectively, indicating that the predicted values exhibit a good correlation with experimental data and that the model is suitable and practicable. The ANOVA for the CCRD experiments (Table 8) revealed that the model exhibits an excellent correlation with the experimental data, with $p<0.01$ for each model and high test statistics *F*-values (41.68 and 912.1 for mycelial growth and erinacine A production, respectively).

Figures 3 and 4 present 3-D response surface and its respective contour plots, which are the graphical representation of Eq. (3) and Eq. (4), respectively. Each plot presents the impact of two independent variables varying within the experimental range on mycelial growth and erinacine A production, while the other two variables were fixed at their respective center point levels. This kind of graphical visualization allows investigation of the relation between experimental levels of each factor and the response, as well as determination of the type of interactions between test variables, which is necessary to establish the optimal composition of the culture medium. The nature of curves in the contour plots provides information about the mutual interactions between variables. The more elliptical shape and the denser distribution of contour lines, the stronger the relation is. The higher density of the contour lines in Figs. 3A and B than that in Fig. 3C suggests that glucose has more significant influence on *H. erinaceum* growth than the other test variables.

Taking into account the data from the 3-D and 2-D response surface plots and equations obtained from multiple regression analysis we can deduce the optimum concentrations of medium components. The model predicts that the maximum mycelial growth (12.91 g/L) was located at $X_1 = 0.94$, $X_2 = -0.01$, $X_3 = 0.14$, and $X_4 = 0.82$. By putting the respective values of X_i into Eq. (1) we have calculated the following real values to be as: glucose 59.44 g/L, casein peptone 20.52 g/L, NaCl 1.03 g/L, and $ZnSO_4 \cdot 7H_2O$ 60.28 mg/L. The predicted erinacine A production (40.3 mg/L) reached its maximum at coded values $X_1 = 1.99$, $X_2 = -1.80$, $X_3 = 1.80$, and $X_4 = 0.42$, corresponding to the actual concentration of glucose 69.87 g/L, casein peptone 11.17 g/L, NaCl 1.45 g/L, and $ZnSO_4 \cdot 7H_2O$ 55.24 mg/L.

For further examination in a bioreactor we chose the culture medium composition for maximum erinacine A production. In this medium the predicted mycelial growth was equal to 11.54 g/L, which is a satisfactory result. To attest the validity of the optimal conditions predicted, four repeated experiments were conducted. The average mycelial growth (11.72 ± 0.41 g/L) and erinacine A production (39.3 ± 1.1 mg/L) were very close to the predicted ones, which indicates that the CCRD model developed in this study is accurate and applicable for predicting the mycelial growth and erinacine A production by submerged cultivation of *H. erinaceum*.

3.5 Cultivation in a bioreactor

Figure 5 presents the changes in mycelial growth and erinacine A production during the cultivation of *H. erinaceum* in a 10-L bioreactor in a medium of the following composition: glucose 69.87 g/L, casein peptone 11.17 g/L, NaCl 1.45 g/L, $ZnSO_4 \cdot 7H_2O$ 55.24 mg/L, and KH_2PO_4 1 g/L; pH 4.5. In the course of the bioprocess the changes in culture medium pH and residual sugar concentration were also controlled. The resulting kinetic parameters were as follows: the specific growth rate $\mu = 0.18 \pm 0.01\ d^{-1}$, the specific production rate of erinacine A $P_{P/X} = 0.479 \pm 0.043$ mg/gd, the specific consumption rate of substrate $Q_{S/X} = 0.049 \pm 0.017$ g/gd, and the yield of erinacine A on substrate $Y_{P/S} = 10.5 \pm 3.2$ mg/g.

Table 7. Estimated regression coefficients for mycelial growth of *H. erinaceum* and erinacine A production.[a)]

	Mycelial growth			Erinacine A production		
	Parameter estimate	SE	*t*-Value	Parameter estimate	SE	*t*-Value
Mean/ intercept	12.48***	0.145	85.9	32.58***	0.242	135
X_1	0.73***	0.051	14.2	1.41***	0.085	16.5
X_2	0.27***	0.051	5.27	−1.18***	0.085	−13.8
X_3	0.16***	0.051	3.18	1.84***	0.085	21.5
X_4	0.17***	0.051	3.32	0.37***	0.085	4.28
X_1^2	−0.34***	0.054	−6.21	−0.17*	0.091	−1.84
X_2^2	−0.28***	0.054	−5.06	0.27***	0.091	3.06
X_3^2	−0.18***	0.054	−3.36	0.54***	0.091	5.92
X_4^2	−0.06	0.054	−1.10	−3.10***	0.091	−34.2
X_1X_2	−0.18***	0.063	−2.81	0.27**	0.105	2.61
X_1X_3	−0.16**	0.063	−2.62	0.03	0.105	0.284
X_1X_4	−0.08	0.063	−1.35	−0.17	0.105	−1.63
X_2X_3	−0.15**	0.063	−2.42	0.30***	0.105	2.94
X_2X_4	−0.11*	0.063	−1.76	−0.04	0.105	−0.403
X_3X_4	0.05	0.063	0.770	−0.13	0.105	−1.24

[a)] X_1 – glucose, X_2 – casein peptone, X_3 – NaCl, X_4 – $ZnSO_4 \cdot 7H_2O$;
* significant at the 10% significance level,
** Significant at the 5% significance level;
***significant at the 1% significance level.

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During the cultivation of *H. erinaceum* in a bioreactor the maximal biomass growth was 13.3 ± 2.6 g/L and was observed on day 15 of cultivation. At the same time depletion of the carbon source in the medium to *ca.* 7% of its initial value was

Table 8. ANOVA for the experimental results of the CCRD.[a)]

	Mycelial growth				Erinacine A			
	SS	df	MS	*F*	SS	df	MS	*F*
Model	41.68	14	2.977	22.41***	912.1	14	65.15	158.2***
X_1	25.45	1	25.45	200.8***	94.78	1	94.78	270.5***
X_2	3.515	1	3.515	27.74***	66.39	1	66.39	189.5***
X_3	1.284	1	1.284	10.13***	162.3	1	162.3	463.1***
X_4	1.397	1	1.397	11.03***	6.417	1	6.417	18.31***
X_1^2	4.883	1	4.883	38.52***	1.189	1	1.189	3.393*
X_2^2	3.245	1	3.245	25.60***	3.270	1	3.270	9.332***
X_3^2	1.433	1	1.433	11.31***	12.27	1	12.27	35.01***
X_4^2	0.1544	1	0.154	1.218	409.5	1	409.5	1169***
X_1X_2	0.9976	1	0.9976	7.871***	2.393	1	2.393	6.829**
X_1X_3	0.8679	1	0.8679	6.848**	0.0282	1	0.0282	0.080
X_1X_4	0.2295	1	0.2295	1.811	0.9282	1	0.9282	2.649
X_2X_3	0.7412	1	0.7412	5.848**	3.032	1	3.032	8.653***
X_2X_4	0.3938	1	0.3938	3.107*	0.0570	1	0.0570	0.163
X_3X_4	0.07508	1	0.0751	0.5924	0.5382	1	0.5282	1.536
Lack of fit	1.505	10	0.1505	1.188	5.896	10	5.896	1.683
Pure error	3.676	29	0.1267		10.16	29	0.3504	
Total SS	46.86	53			928.1	53		

[a)] X_1 – glucose, X_2 – casein peptone, X_3 – NaCl, X_4 – $ZnSO_4 \cdot 7H_2O$; *significant at the 10% significance level; ** significant at the 5% significance level; ***significant at the 1% significance level. df, degrees of freedom; MS, mean square; SS, sum of squares.

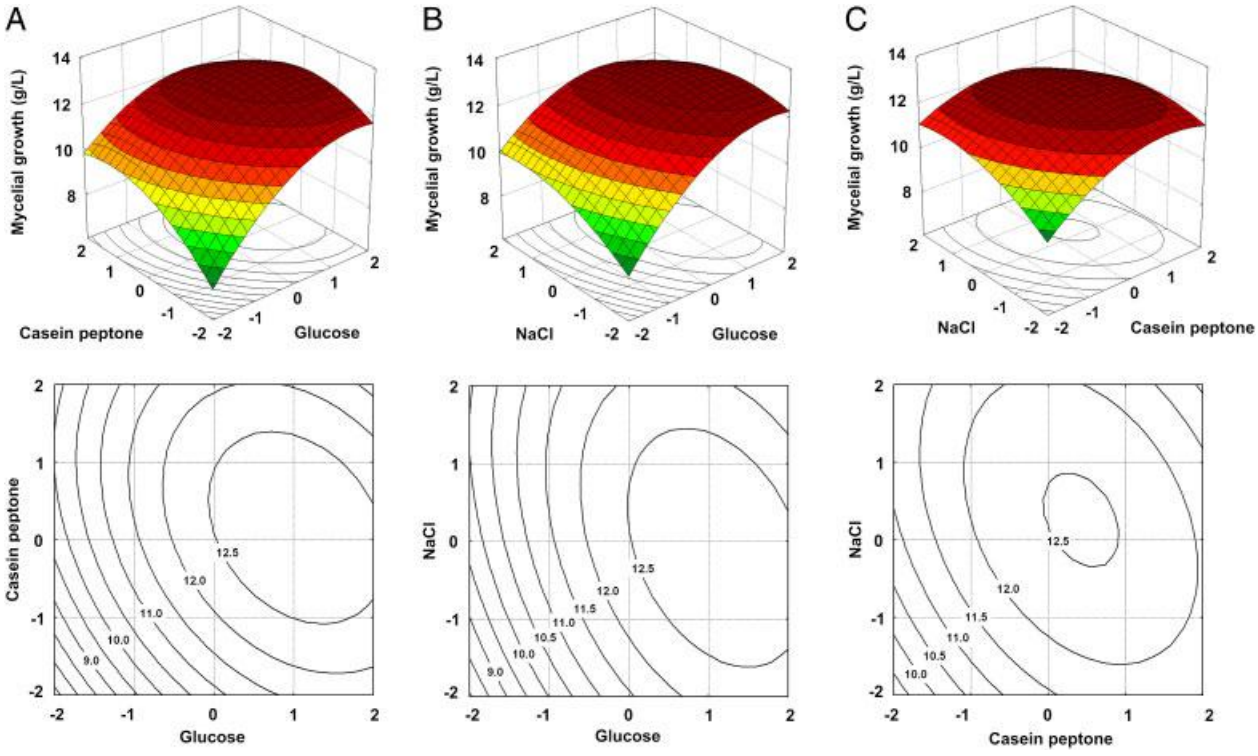

Figure 3. Response surface and corresponding contour plots showing the combined effect of (A) glucose and casein peptone, (B) glucose and NaCl, and (C) casein peptone and NaCl on mycelial growth of *H. erinaceum*. Other variables were fixed at their respective center point levels.

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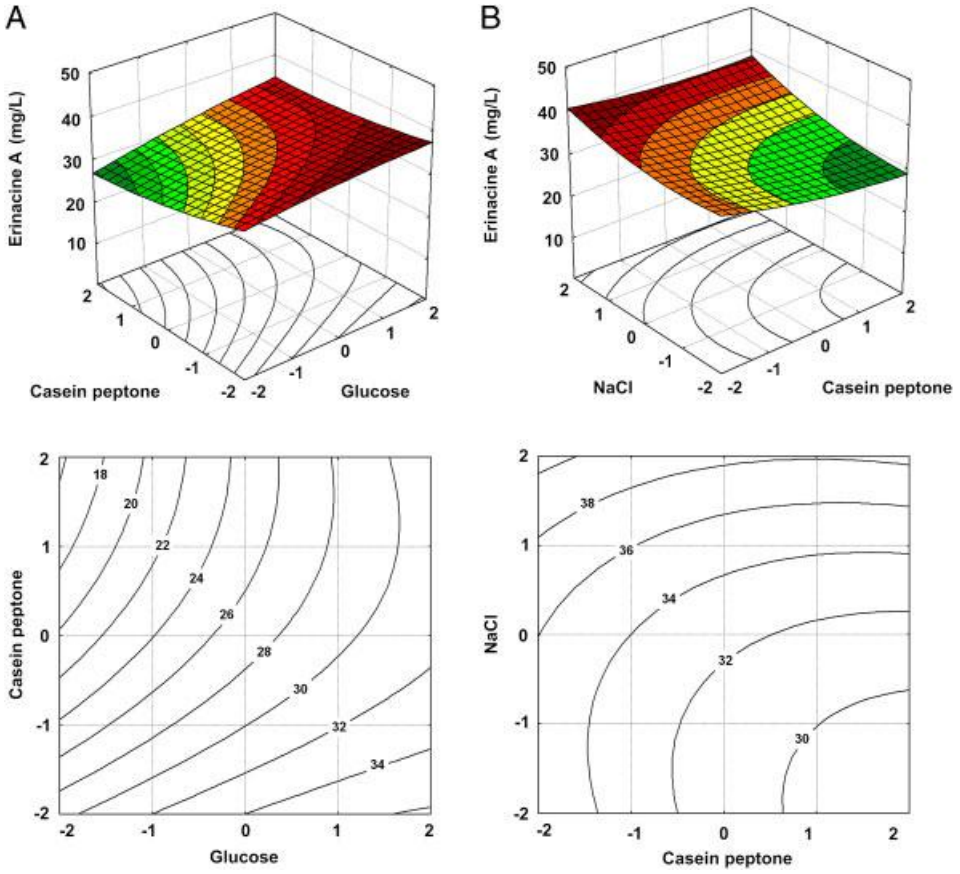

Figure 4. Response surface and corresponding contour plots showing the combined effect of (A) glucose and casein peptone and (B) casein peptone and NaCl on erinacine A production. Other variables were fixed at their respective center point levels.

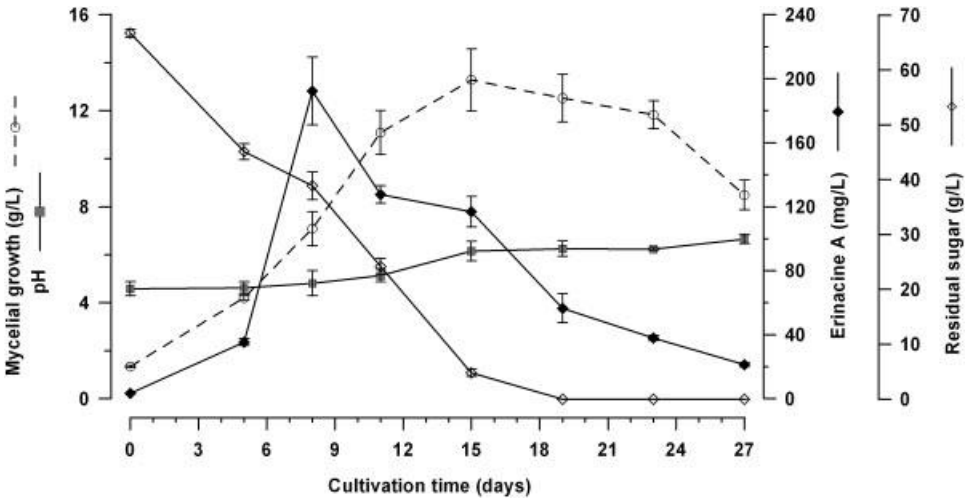

Figure 5. Time courses of mycelial growth and erinacine A production of *H. erinaceum* in 10-L bioreactor in optimized medium. All experimental data are values of mean ± SD of triplicate determinations.

noticed. The decrease of biomass yield on the 19th day of cultivation due to the depletion of glucose in the culture medium indicated that cell metabolism changed from the logarithmic growth phase to the stationary phase. The culture medium pH increased slowly to the end of cultivation from its initial value of 4.58 ± 0.59 to 6.65 ± 0.39. The lowering of the culture medium pH due to secretion of organic acids during the cultivation has often been observed in submerged cultures of mushrooms [16, 33]. However, some authors found a gradual pH increase during the cultivation of higher mushrooms in a bioreactor [14, 27]. As shown in Fig. 5, an extensive production of erinacine A started after 5 days of cultivation, when the glucose concentration decreased below 45 g/L, and the biomass concentration was 5.3 times greater than at the beginning of cultivation. This phenomenon can be explained by the repressive effect of high glucose concentration on

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erinacine A biosynthesis [26]. It was observed that the production of erinacine A attained a maximum of 192 ± 42 mg/L on day 8 of cultivation and then decreased slightly. It seems that erinacine A is not a non-growth-associated metabolite as it was regarded [3–5, 9] but a growth-associated metabolite that is easily transformed to another cyathan metabolite [8, 34]. Thus the optimal time to obtain the highest production of erinacine A during cultivation of *H. erinaceum* in a bioreactor is 8 days.

4 Concluding remarks

During this study a new method for quantitative erinacine A determination was developed and validated. It was proved that the method is reproducible, precise, accurate, and specific and can be applied routinely in research laboratories. In the next step of our investigations the CCRD experiment combined with RSM analysis was successfully applied to optimization of erinacine A biosynthesis by cultured mycelium of *H. erinaceum*. The fermentation kinetics in a bioreactor showed that the production of erinacine A proceeded most effectively during exponential phase. The proposed composition of culture medium can be used for hyperproduction of erinacine A on a bioreactor scale as well as for effective biosynthesis of *H. erinaceum* biomass. The techniques of culture medium optimization presented in this report can be applied for optimization of submerged cultures of other mushrooms, especially those producing valuable bioactive metabolites.

Conflict of interest

The authors have declared no conflict of interest.

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